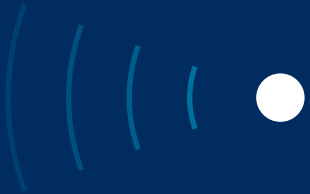
Honk-Sense

A car that understands Indian roads by ear

Directional alerts for the horns and sirens that matter.
Silence for everything else.



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HCoE Mobility Innovation Ideathon 2026 ▪ Topic 1: India-Smart Interaction ▪ Track A

The Problem: Loud Roads, Deaf Cabins

>90 dB
daily honking exposure for Mumbai commuters (IIT Bombay study)

>100 dBA
two-wheeler horns measured at 3 m in Mumbai (MPCB)

25–35 min
for ambulances to reach accident scenes on core national highways

- **Alert fatigue:** drivers hear so many honks that they tune *all* of them out, including the few that signal real danger.
- **The cabin paradox:** better insulation, closed windows, AC and music make modern cabins quieter, so sirens and nearby horns are heard *later*.
- **No sense of direction:** even when a driver hears a siren, it is hard to tell where it is coming from in dense traffic.
- **Ambulances lose time:** drivers notice them late and do not know which way to move.

What the Data Says

4.8 lakh

road accidents in 2023; 1.72 lakh deaths

36%

rear-end (21%) + side (15%) collisions

What We Claim

- **14–31% improvement** from awareness systems in similar situations: blind-spot monitoring cut lane-change crashes by 14% (IIHS), and side blind-zone alerts cut lane changes made without a mirror check by 31%.
- We do not claim a number for Honk-Sense yet.

Why Accidents Still Happen Despite Existing Systems

The problem: important sounds outside the car do not reach the driver in a usable form: on time, with a direction, and separated from the noise.

- **Mirrors depend on attention.** Drivers do not check them, or check and fail to register: 75% of lane-change crashes involve this recognition failure.
- **Blind-spot systems are not universal.** Radar BSM is mostly on upper variants, and tuned for orderly lanes, not weaving two-wheelers.
- **Every sensor needs line of sight.** Nothing sees around a bus, a truck or a hairpin bend.
- **Proximity sensors are short-range.** Ultrasonic sensors reach about 2 m and are made for parking.
- **Sirens lose to the cabin.** Sealing, AC and music cut them down; drivers placed a siren correctly only 26% of the time.
- **Too many horns means no horn.** Habituation makes the meaningful honk invisible.

Gap: turning outside sound into a short, directional, filtered cue inside the car.

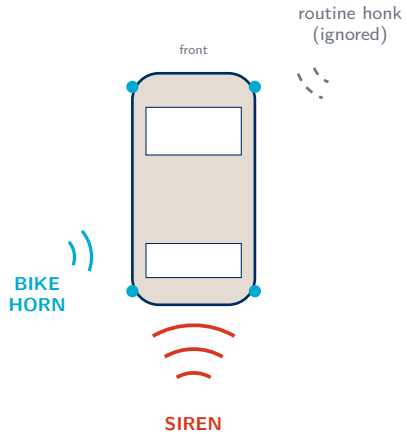
Sources: FMCSA (lane-change recognition failure); US DOT siren audibility study; TI SPRY331 (ultrasonic range); SAE 2026-26-0035 (ADAS in India). Full links on References (2).

Honk-Sense in One Slide

The idea

Exterior microphones and an on-board AI **classify every road sound**, **locate where it comes from**, and decide whether the people inside need to know.

- **Ambulance behind?** Music dips, a rear chime plays: *"Ambulance behind, keep left"*.
- **Bike honking in the blind spot?** A chime from that side, plus an arrow.
- **Routine honking in a jam? Nothing.** The cabin stays calm.
- **Fleet:** cars that hear a siren warn Hyundai cars ahead.



Comparison with Existing Technologies

Situation	Mirrors	Dashcam	Ultra-sonic	Radar BSM	Camera ADAS	Honk-Sense
Ambulance hidden in traffic	–	–	–	–	–	✓
Horn behind a blind turn	–	–	–	–	–	✓
Which side a horn came from	–	–	–	–	–	✓
Honking bike in blind spot	∼	–	–	✓	–	✓
Silent car in blind spot	∼	–	–	✓	–	–
Parking, low-speed obstacles	∼	–	✓	–	∼	–
Lane departure, car ahead	–	–	–	–	✓	–
Evidence after a crash	–	✓	–	–	–	–

✓ does it ∼ partly – cannot

Other systems need line of sight. Only Honk-Sense brings a horn's intent into the cabin, with a direction.

The Horn as a Language on Indian Roads

On Indian roads, the horn is a language

- **"I'm here"**: a bike announcing itself in your blind spot
- **"I'm overtaking"**: trucks invite it with *"Horn OK Please"* painted on the back
- **"Careful, blind turn"**: hill roads have *"Blow Horn"* signs
- **"Move"** or **"signal is green"**: impatience in jams
- **Siren**: give way now

How drivers respond

- **Habituation:** so many honks that *all* are ignored, including the important ones
- **Sealed cabins:** windows up, AC and music on, so outside sounds arrive late
- **Direction confusion:** a siren is heard, but not from where; drivers freeze or move the wrong way
- **Hearing gaps:** elderly and hearing-impaired drivers rely on mirrors alone

The car should understand this language the way an experienced Indian driver does, and **translate only what matters**.

Why This Answers India-Smart Interaction

Question

How can vehicles understand and interact with Indian users more naturally?

Honk-Sense extends "understanding" from the **user's voice** to the **user's environment**:

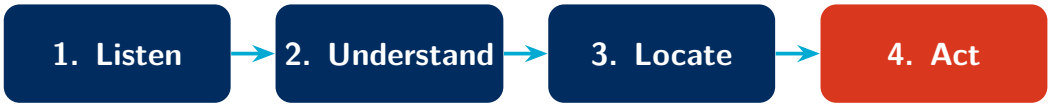
- understands India-specific sounds: two-wheeler horns, auto-rickshaw horns, pressure horns, Indian siren patterns;
- interacts naturally: the right speaker, the right moment, in the user's language.

Voice assistants understand what you say. Honk-Sense understands what the road is saying.

A Drive with and without Honk-Sense

Moment	Today	With Honk-Sense
Stuck in a honking jam	Constant noise and stress; every honk ignored	**Calm cabin.** Nothing is announced
Bike honks during a lane change	A horn somewhere; driver unsure where	Chime from the **left** speakers and a left arrow
Ambulance 200 m behind, music on	Noticed late; driver freezes	Music dips: *"Ambulance behind, keep left"*
Ambulance 1 km behind	No idea it is coming	Fleet notice: *"Ambulance approaching, about 1 minute"*
Truck overtaking on a highway	Pressure horn startles the family	Soft cue: *"Truck overtaking on the right"*
Hearing-impaired driver	Misses sirens entirely	Arrow on the cluster and a seat vibration

The Experience in Four Steps



Silence is a feature

The car stays **quiet** for everyday honks and speaks up **only when it matters**.

Use Cases (1/3): Emergencies and Horn Direction



U1. Hidden ambulance

An ambulance is behind a bus; music on, windows up. Honk-Sense shows *"Ambulance, behind left, approaching"*, dips the music and suggests which way to move.



U2. Which horn, and where?

Horns from every side; one is a bike at your rear-left as you turn left. Honk-Sense shows only the horns that are close and directed at you, with an arrow, and ignores the rest.

Photos: Wikimedia Commons. U1: Biswarup Ganguly (CC BY 3.0); U2: Amityadav8 (public domain).

Use Cases (2/3): Blind Spots and Blind Turns



U3. Honk in the blind spot

A bike or truck in your blind spot honks before you change lanes; most cars in India have no radar blind-spot monitor. Honk-Sense: *"Horn, right rear"* as you signal.



U4. Around the bend

A bus honks at a hairpin bend before it is in view. Honk-Sense warns *"Vehicle ahead, around the bend"* seconds earlier.

Cameras and radar cannot see around the corner; sound can.

Photos: Wikimedia Commons. U3: Justin Morgan (CC BY-SA 2.0); U4: Rohini (CC BY-SA 3.0).

Use Cases (3/3): A Calmer Cabin



Photo: Wikimedia Commons. Divya Chinnasamy (CC0).

U5. Fewer interruptions

- Peak-hour traffic, with street noise above 90–100 dB.
- Drivers hear so many horns that they stop reacting to all of them.
- Less stress, and less alarm fatigue.

In-Cabin Experience: Three Alert Tiers

Tier	Triggered by	What the driver gets	What passengers get
0. Silent	Routine honks, distant horns, sounds ahead in a jam	Nothing (logged only)	Nothing
1. Aware	Horn very close at the side or rear; truck overtaking	Short directional chime from the matching speaker; arrow icon on the cluster	Nothing
2. Urgent	Siren approaching; horn plus fast approach during a lane change	Music ducked, spoken cue in the driver's language, arrow and suggested action	Rear-screen banner: *"Ambulance approaching"*

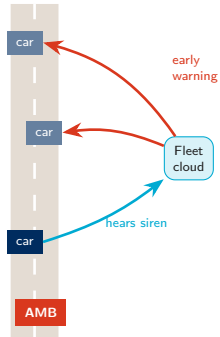
Spatial audio cue
The chime plays from the speaker nearest the sound source, so the driver's ears point the right way *before* they read anything.

Accessibility
Hearing-impaired mode: visual and seat-haptic alerts only. Volume, languages and tiers are configurable per driver profile.

Fleet Layer: Clearing the Way for Ambulances

1. A Hyundai car hears a siren and estimates its direction of travel.
2. It shares only four facts with Hyundai's connected-car cloud: a siren was heard, where, which way it is heading, and when.
3. The cloud confirms the event when several cars report it, which filters false alarms.
4. Hyundai cars **ahead on the same road** get an early notice: *"Ambulance approaching from behind in about 1 minute. Prepare to keep left."*

Future: with city emergency services, the same channel could carry official ambulance routes.



What Makes It Possible

Only one new part

4 small exterior microphones. Low-cost parts, and automotive-grade versions already exist for other exterior-microphone features.

Everything else is already in the car

- Infotainment computer runs the AI
- Cabin speakers play the directional cue
- Instrument cluster shows the arrow
- Connected-car modem shares sirens

Runs inside the car, offline

- **Fast:** about 1 second from sound to alert, while a siren is audible hundreds of metres away
- **Private:** no audio is stored or leaves the car
- **Works in dead zones:** no network needed except for the optional fleet relay
- **Upgradeable:** improves through over-the-air updates

A technical deep-dive is available on request.

What Exists, and What Is New

Prior work	What it does	Honk-Sense goes further
Siren detection and localisation patents (e.g. Motional)	Detect siren, estimate bearing, mainly for autonomous driving	Built for *human* occupants, with tiered alerts
Sound-direction cabin feedback patent (US10351148B2)	Haptic or audio cue for direction	Adds India-specific classes and *silence by default*
Horn/siren detection research (e.g. Forum Acusticum 2025)	Compares interior and exterior mics; stationary car	Works while driving, in wind and rain
Indian horn-detection projects	Noise policing in silence zones	In-vehicle driver assistance

Our three novel elements

1. **India-trained sound understanding**: two-wheeler, auto, pressure horns and Indian sirens.
2. **Alert-fatigue-first design**: deciding what *not* to announce is the core feature.
3. **Fleet siren relay**: cars that hear an ambulance warn the cars ahead of it. (No production car in India found with this, Oct 2026.)

Value for Hyundai

For customers

- Calmer, less stressful city driving
- Safer lane changes around two-wheelers
- Inclusive: helps hearing-impaired and elderly drivers

For society

- Ambulances get through faster
- Anonymised honk-density maps could help cities target noise hot-spots

For Hyundai

- A visible, India-first feature, easy to demonstrate in showrooms
- Mostly software on existing hardware: scales across the line-up
- Fits connected-car subscriptions (fleet siren relay)

Mostly software, on hardware the car already has.

Roadmap

Ideathon (now)

Phase 1

Concept, usage scenarios and experience design (this submission)

Phase 2

Driver survey; proof of concept on a mic array; Indian road recordings

Phase 3

Integration on a Hyundai IVI platform; automotive mics; user study

Phase 4

Feature in Indian models; fleet siren relay; partner with emergency services

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Ambulance, Kolkata: Biswarup Ganguly (CC BY 3.0) ▪ Pune two-wheeler traffic: Amityadav8 (public domain) ▪ Blow Horn: Justin Morgan (CC BY-SA 2.0) ▪ Hairpin bend, Western Ghats: Rohini (CC BY-SA 3.0) ▪ Traffic in Bangalore: Divya Chinnasamy (CC0)

Thank you

Honk-Sense: the car hears what matters, so you don't have to

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